NASA Space Apps Challenge 2026

Challenge Brief: Interplanetary Survival Map

1. Challenge Overview

Create interactive maps and survival tools for future Mars explorers.

NASA’s current and past Mars missions have collected extensive scientific data about the planet. However, these datasets are not yet integrated in ways that fully support the planning of specific human surface excursions.

Existing platforms, such as NASA/JPL’s Mars Trek and the USGS–NASA Planetary Geologic Mapping Program, provide global and regional views of Mars. This challenge focuses on developing a more localized and mission-oriented mapping tool that helps future astronauts understand their surroundings, plan scientific activities, navigate terrain, and make informed safety decisions.

2. Background

Human explorers can conduct scientific activities and make real-time observations that robotic systems cannot fully replicate. Through critical thinking and contextual understanding, astronauts can connect scientific measurements, respond to unexpected conditions, and make decisions in complex environments.

NASA's robotic missions to Mars have significantly expanded our understanding of the planet. However, sending humans to Mars will introduce much greater engineering, operational, and safety challenges.

A human Mars mission will require reliable tools that combine scientific information, environmental observations, terrain analysis, and navigation support. These tools should help astronauts plan and execute surface excursions while understanding the conditions and potential risks of their surroundings.

3. Main Objectives

Develop a **map, tool, or interactive interface** that a Mars astronaut could use during a surface exploration mission, using NASA data and available tools.

The solution should help astronauts:

- Identify scientific objectives and select exploration locations.
- Explore areas previously studied by NASA missions or investigate less-explored regions.
- Combine different types of planetary data to understand the environment.

- Plan surface routes while considering terrain conditions and potential obstacles.
- Access useful information for navigation, scientific exploration, and astronaut safety.

4. Data Integration Requirements

Consider incorporating at least three different types of Mars-related data, layered or combined to provide useful context for surface navigation.

Potential data categories include:

- **Optical:** Surface imagery and visual observations.
- **Thermal:** Surface temperature and thermal information.
- **Spectral:** Mineralogical and compositional information.
- **Topographic:** Elevation, slope, and terrain structure.
- **Geophysical:** Subsurface and geological information.
- **Atmospheric:** Weather and atmospheric conditions.
- **Radiation:** Radiation environment and space-weather information.
- **Chemical:** Surface composition and chemical observations.

The selected datasets should contribute meaningful scientific or operational information to the exploration interface.

5. Astronaut Safety and Navigation

Astronaut safety is NASA's highest priority.

The application should consider information that could help astronauts move safely across the Martian surface, including:

- Terrain elevation, slope, and surface obstacles.
- Route distance and estimated travel requirements.
- Potentially hazardous terrain and environmental conditions.
- Radiation and atmospheric information, where available.
- The mobility limitations of astronauts wearing spacesuits and carrying equipment.
- Suitable routes between exploration points.
- Potential safe stops and alternative routes, where feasible.

The interface should help astronauts understand the conditions along a planned route before and during an exploration activity.

6. Potential Considerations

The challenge allows flexibility in how the solution is designed and presented.

Consider the following possibilities:

- The interface could function as a spacecraft or habitat planning display, a portable tablet, a spacesuit heads-up display (HUD), or another suitable system.
- The solution may be a static map, an interactive digital application, or a tool capable of generating maps for different locations and datasets.
- Existing planetary mapping systems and maps from other NASA missions may provide useful design inspiration.
- Data may represent historical observations, predicted future conditions, or relatively stable planetary information such as topography and geology.
- Surface imagery from NASA's Mars missions could help astronauts understand what they may encounter at a destination.
- Robotic systems or rovers could support astronauts by transporting equipment, collecting information, or assisting with exploration.
- Astronauts may work in pairs to conduct scientific activities and improve mission safety.

7. Suggested Technical Direction for Our Implementation

The following is our proposed implementation approach, not a mandatory NASA requirement.

A. Multi-Location Mars Exploration

Support all 10 Martian locations already included in our application. Each location should use its corresponding terrain data and available scientific information.

B. Interactive Traverse Planning

Allow users to draw a route across the selected Martian location. Analyze the route using the corresponding terrain elevation data.

Calculate:

- Total route distance.
- Elevation profile.
- Terrain slope along route segments.
- Total elevation gain.
- Estimated travel duration.
- Estimated energy requirements.
- Route feasibility based on documented mobility or rover constraints.

C. Terrain and Hazard Analysis

Highlight steep or potentially hazardous route sections. Explain why a route may fail a specified constraint and identify alternative routes when possible.

D. Survival Card

Generate a route-dependent mission summary containing:

- Estimated mission duration.
- Estimated energy requirements and reserves.
- Maximum route slope.
- Safe stopping locations, where supported by data.
- Identified terrain hazards.
- Suggested contingency actions.

E. Route Feasibility Demonstration

A demonstration could involve selecting a Martian location, drawing a route, displaying its elevation profile, highlighting steep sections, and calculating estimated travel requirements.

If a route exceeds a documented slope limit, the system should identify the problematic segment and explain the reason. The user can then modify the route or examine an alternative.

8. Suggested NASA Data Sources and Technologies

- **NASA/JPL Mars Trek:** Mars imagery, elevation information, mapping layers, and downloadable regional products.
- **NASA Planetary Data System (PDS):** Scientific mission datasets and authoritative archival references.
- **USGS–NASA Planetary Geologic Mapping Program:** Planetary geological maps and related information.
- **SpiceyPy / SPICE kernels:** Optional support for calculating spacecraft or communication geometry when appropriate kernels and data are available.

9. Technical Planning Notes

A possible implementation architecture:

1. Obtain suitable Mars imagery and regional elevation data.
2. Prepare and document the datasets, including their sources, spatial coverage, resolution, and units.
3. Display the selected region using an interactive Mars map.
4. Allow the user to draw a route.
5. Sample elevation values along the route.
6. Calculate distance, slope, elevation gain, travel time, and estimated energy requirements.
7. Display an elevation profile, hazard overlay, and route-dependent Survival Card.
8. Optionally calculate a least-cost alternative route and contingency route.

All model parameters and assumptions should be visible to users. Estimated values must be clearly distinguished from actual NASA observations and validated measurements.

10. Evaluation Priorities

The solution should prioritize:

1. **Scientific credibility:** Use authentic, properly documented planetary data.
2. **Mission plausibility:** Make assumptions, constraints, and calculations transparent.
3. **Navigation usefulness:** Help users understand and evaluate routes across Martian terrain.
4. **Astronaut safety:** Communicate potential risks and relevant environmental information.
5. **Data integration:** Combine multiple types of scientific information into a useful interface.
6. **Practical functionality:** Demonstrate a working exploration-planning workflow rather than relying only on visual simulations.

Overall Goal: Build a scientifically grounded, interactive Mars exploration and survival-planning system that integrates planetary data, terrain-aware route analysis, scientific context, and astronaut safety support for future human surface missions.